

Project 1.1 : Implementing Jenkins Pipeline as code Using Jenkinsfile

Project Tasks:

Task 1 - Prepare the GitHub Repository :

<https://github.com/SnehalShinde11/Jenkins-Maven-Project/tree/main>

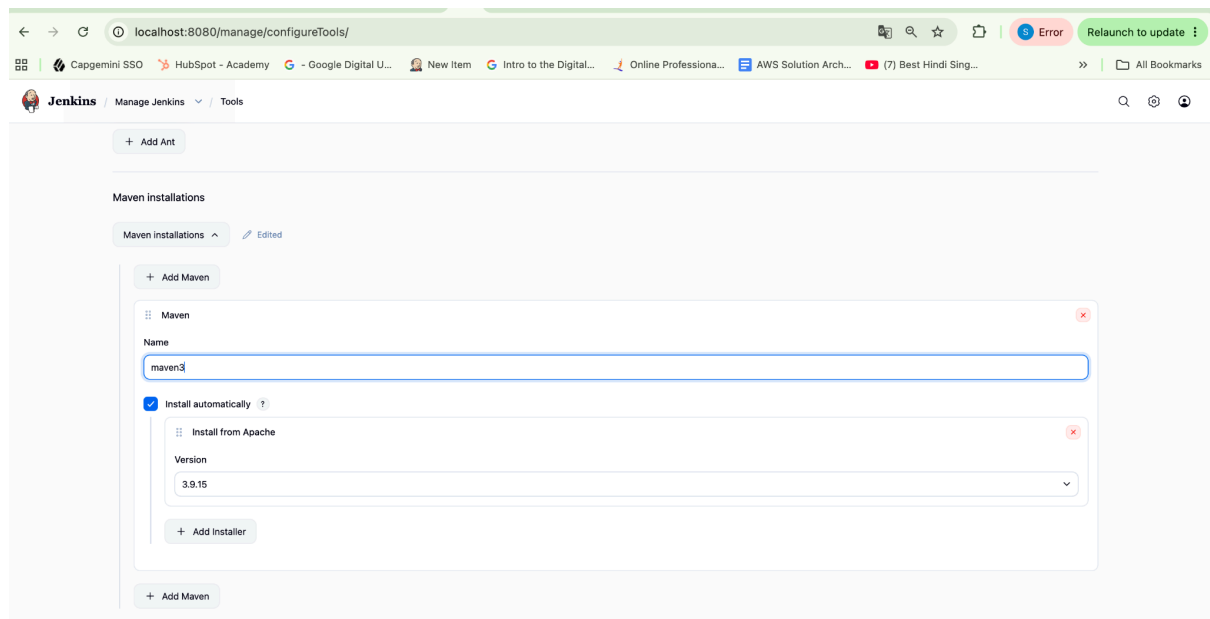
Create a Java Maven repository containing:

- Java source code
- pom.xml
- README documentation

Task 2- Create Jenkinsfile.

-Ref Repo: <https://github.com/SnehalShinde11/Jenkins-Maven-Project/tree/main>

Before Task 3 : Make sure to add maven Configuration under tools section in Jenkins manage jenkins section as below :



Task 3 - Create Jenkins Pipeline Job

In Jenkins:

New Item → Pipeline

Configure:

Pipeline Definition:

Pipeline script from SCM

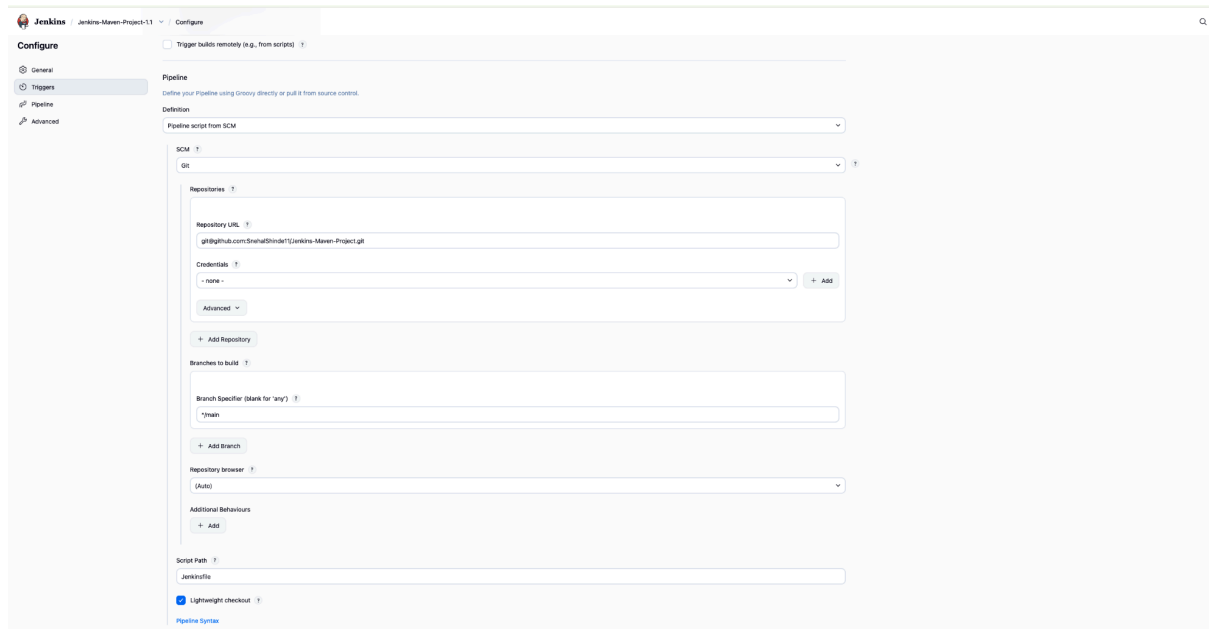
Repository:

GitHub repository URL

Script Path:

Jenkinsfile

Jenkins will now read the pipeline configuration from the repository.



Task 4/5: Run the Pipeline , Verify Pipeline Execution.

Deliverables

1. GitHub repository link : `git@github.com:SnehalShinde11/Jenkins-Maven-Project.git`
2. Screenshot of Jenkins pipeline configuration.

Jenkins / Jenkins-Maven-Project-1.1 / Configure

Configure

General

Triggers

Pipeline

Advanced

Trigger builds remotely (e.g., from scripts)

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script from SCM

SCM

Git

Repositories

Repository URL

github.com:5neph0st0ider/Jenkins-Maven-Project.git

Credentials

none

Add

Advanced

Add Repository

Branches to build

Branch Specifier (blank for 'any')

*/main

Add Branch

Repository browser

Auto

Additional Behaviours

Add

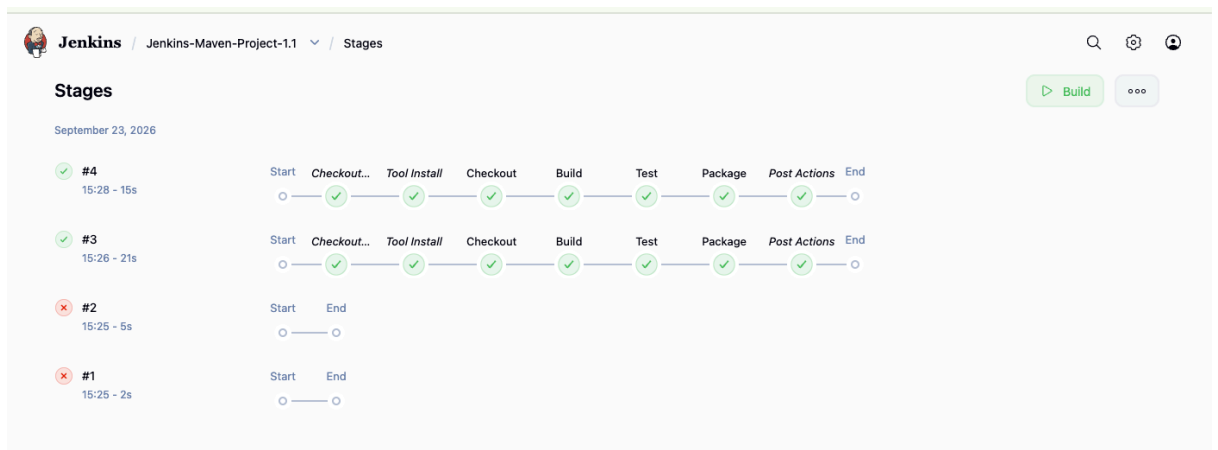
Script Path

Jenkinsfile

Lightweight checkout

Pipeline Syntax

3.Screenshot of pipeline stage view



Jenkins / Jenkins-Maven-Project-1.1 / #4

Status ✓ #4 (Sep 23, 2026, 3:28:39 PM) Add description Keep this build forever

Build Artifacts
[pipeline-ci-demo-1.0.0-SNAPSHOT.jar](#) 2.31 KiB [view](#)

Started by user [admin-jenkins](#)

This run spent:

- 9 ms waiting;
- 15 sec build duration;
- 15 sec total from scheduled to completion.

Revision: d27ace07a7259ce4de609215633830bc414ad283
Repository: [git@github.com:SnehalShinde11/Jenkins-Maven-Project.git](#)

- refs/remotes/origin/main

No changes.

Left sidebar:

- </> Changes
- Console Output
- Edit Build Information
- Delete build '#4'
- Timings
- Git Build Data
- See Fingerprints
- Pipeline Overview
- Edit build information
- Restart from Stage
- Replay
- Pipeline Steps
- Workspaces
- Previous Build

- Screenshot of Jenkins console output

Jenkins / Jenkins-Maven-Project-1.1 / #4

Console Download Copy View as plain text

```

Started by user admin-jenkins
Obtained Jenkinsfile from git git@github.com:SnehalShinde11/Jenkins-Maven-Project.git
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /Users/snehal/.jenkins/workspace/Jenkins-Maven-Project-1.1
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Checkout SCM)
[Pipeline] checkout
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
No credentials specified
> git rev-parse --resolve-git-dir /Users/snehal/.jenkins/workspace/Jenkins-Maven-Project-1.1/.git # timeout=10
Fetching changes from the remote Git repository
> git config remote.origin.url git@github.com:SnehalShinde11/Jenkins-Maven-Project.git # timeout=10
Fetching upstream changes from git@github.com:SnehalShinde11/Jenkins-Maven-Project.git
> git --version # timeout=10
> git --version # 'git version 2.58.1 (Apple Git-155)'
> git fetch --tags --force --progress -- git@github.com:SnehalShinde11/Jenkins-Maven-Project.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision d27ace07a7259ce4de609215633830bc414ad283 (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f d27ace07a7259ce4de609215633830bc414ad283 # timeout=10
Commit message: "Remove JDK specification from Jenkinsfile"
> git rev-list --no-walk d27ace07a7259ce4de609215633830bc414ad283 # timeout=10
[Pipeline] }
[Pipeline] // stage
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Tool Install)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] }
[Pipeline] // stage
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Checkout)
[Pipeline] tool

```

Left sidebar:

- Status
- </> Changes
- Console Output
- Edit Build Information
- Delete build '#4'
- Timings
- Git Build Data
- See Fingerprints
- Pipeline Overview
- Edit build information
- Restart from Stage
- Replay
- Pipeline Steps
- Workspaces
- Previous Build

```
Jenkins Jenkins-Maven-Project-1.1 #4
[INFO] skip non existing resourceDirectory /Users/sneha/.jenkins/workspace/Jenkins-Maven-Project-1.1/src/main/resources
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ pipeline-ci-demo ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ pipeline-ci-demo ---
[INFO] skip non existing resourceDirectory /Users/sneha/.jenkins/workspace/Jenkins-Maven-Project-1.1/src/test/resources
[INFO]
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ pipeline-ci-demo ---
[INFO] No sources to compile
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ pipeline-ci-demo ---
[INFO] No tests to run.
[INFO]
[INFO] --- jar:3.4.2:jar (default-jar) @ pipeline-ci-demo ---
[INFO] Building jar: /Users/sneha/.jenkins/workspace/Jenkins-Maven-Project-1.1/target/pipeline-ci-demo-1.0.0-SNAPSHOT.jar
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 0.409 s
[INFO] Finished at: 2026-09-23T15:28:54+05:30
[INFO]
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] archiveArtifacts
Archiving artifacts
Recording fingerprints
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Project 1.2: Implementing Parallel CI Pipeline in Jenkins

Task 1: Prepare the Github repo : Using the same repo.

Task 2: Create the Jenkins Pipeline Job.

The screenshot shows the Jenkins configuration interface for a project named 'Jenkins-Maven-Project-1.2'. The left sidebar contains navigation links for 'General', 'Triggers', 'Pipeline', and 'Advanced', with 'Pipeline' currently selected. The main configuration area is titled 'Configure' and includes a 'Definition' dropdown set to 'Pipeline script from SCM'. Below this, the 'SCM' section is set to 'Git'. The 'Repositories' section contains a single repository with the URL 'git@github.com:SnehalShinde11/Jenkins-Maven-Project.git', credentials set to 'none', and an 'Advanced' dropdown. There is an 'Add Repository' button. The 'Branches to build' section has a 'Branch Specifier (blank for 'any')' set to '*/main' and an 'Add Branch' button. The 'Repository browser' is set to '(Auto)'. The 'Additional Behaviours' section has an 'Add' button. At the bottom, the 'Script Path' is set to 'Jenkinsfile2'. A small 'Link to Jenkinsfile2' link is visible at the bottom left of the configuration area.

Task 3: Create Jenkinsfile with Parallel Stages
Jenkinsfile2 here .
By default in jenkinsfile we have checkout to scm.

Task4: Execute the Pipeline

Jenkins / Jenkins-Maven-Project-1.2

Status

- Changes
- Build Now
- Delete Pipeline
- Configure
- Rename
- Stages
- Pipeline Syntax

Jenkins-Maven-Project-1.2

Permalinks

- Last build (#2), 1 min 6 sec ago
- Last stable build (#2), 1 min 6 sec ago
- Last successful build (#2), 1 min 6 sec ago
- Last completed build (#2), 1 min 6 sec ago

Builds

Filter

Today

- #2 4:34 PM
- #1 4:32 PM

Task5: Verify Parallel Execution

Jenkins / Jenkins-Maven-Project-1.2 / #2 / Stages

#2

Started by admin-jenkins | Started 42 sec ago | Queued 3 ms | Took 12 sec

Start | Checkout SCM | Tool Install | Build | Parallel Tests | End

Parallel Tests

- Unit Tests
- Integration Tests
- Code Analysis

Code Analysis

Use a tool from a predefined Tool Installation: maven3

Fetches the environment variables for a given tool in a list of "FOO=bar" strings suitable for the withEnv step.

mvn pmd:pmd

```

0 + mvn pmd:pmd
1 [INFO] Scanning for projects...
2 [INFO]
3 [INFO] ----- com.example:pipeline-ci-demo -----
4 [INFO] Building pipeline-ci-demo 1.0.0-SNAPSHOT
5 [INFO] from pom.xml
6 [INFO] ----- [ jar ] -----
7 [INFO]
8 [INFO] --- pmd:3.28.0:pmd (default-cli) @ pipeline-ci-demo ---
9 [WARNING] Unable to locate Source XRef to link to -- DISABLED
10 [INFO] PMD version: 7.17.0
11 [INFO] Rendering content with org.apache.maven.skis:maven-fluido-skin:jar:2.0.0-M9 skin
12 [INFO]
13 [INFO] BUILD SUCCESS
14 [INFO]
15 [INFO] Total time: 1.526 s
16 [INFO] Finished at: 2026-09-23T16:34:15+05:30
17 [INFO]

```

Deliverables:

Github repository link :<https://github.com/SnehalShinde11/Jenkins-Maven-Project>

Screenshot of jenkins pipeline Configuration

Jenkins

Jenkins-Maven-Project-1.2

Configure

Configure

General

Triggers

Pipeline

Advanced

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script from SCM

SCM

Git

Repositories

Repository URL

git@github.com:SnehalShinde11/Jenkins-Maven-Project.git

Credentials

- none -

Add

Advanced

+ Add Repository

Branches to build

Branch Specifier (blank for 'any')

*/main

+ Add Branch

Repository browser

(Auto)

Additional Behaviours

+ Add

Script Path

Jenkinsfile2

☒ Lightweight checkout

[Pipeline Syntax](#)

Screenshot of jenkins console logs.

Jenkins

Jenkins-Maven-Project-1.2

#2

Status

Changes

Console Output

Edit Build Information

Delete build '#2'

Timings

Git Build Data

Pipeline Overview

Edit build information

Restart from Stage

Replay

Pipeline Steps

Workspaces

Previous Build

Console

Download

Copy

View as plain text

Started by user admin-jenkins

Obtained Jenkinsfile2 from git git@github.com:SnehaShinde11/Jenkins-Maven-Project.git

[Pipeline] Start of Pipeline

[Pipeline] node

Running on Jenkins in /Users/sneha/.jenkins/workspace/Jenkins-Maven-Project-1.2

[Pipeline] {

[Pipeline] stage

[Pipeline] { (Declarative: Checkout SCM)

[Pipeline] checkout

Selected Git installation does not exist. Using Default

The recommended git tool is: NONE

No credentials specified

> git rev-parse --resolve-git-dir /Users/sneha/.jenkins/workspace/Jenkins-Maven-Project-1.2/.git # timeout=10

Fetching changes from the remote Git repository

> git config remote.origin.url git@github.com:SnehaShinde11/Jenkins-Maven-Project.git # timeout=10

Fetching upstream changes from git@github.com:SnehaShinde11/Jenkins-Maven-Project.git

> git --version # timeout=10

> git --version # 'git version 2.50.1 (Apple Git-155)'

> git fetch --tags --force --progress -- git@github.com:SnehaShinde11/Jenkins-Maven-Project.git +refs/heads/*:refs/remotes/origin/* # timeout=10

> git rev-parse refs/remotes/origin/main^{commit} # timeout=10

Checking out Revision 4d6721066702bc6c2cf3a43ff9f91f790af70da3 (refs/remotes/origin/main)

> git config core.sparsecheckout # timeout=10

> git checkout -f 4d6721066702bc6c2cf3a43ff9f91f790af70da3 # timeout=10

Commit message: "Add Maven tool configuration to Jenkinsfile"

> git rev-list --no-walk 4d6721066702bc6c2cf3a43ff9f91f790af70da3 # timeout=10

[Pipeline] }

[Pipeline] // stage

[Pipeline] withEnv

[Pipeline] {

[Pipeline] stage

[Pipeline] { (Declarative: Tool Install)

[Pipeline] tool

[Pipeline] envVarsForTool

[Pipeline] }

[Pipeline] // stage

[Pipeline] withEnv

[Pipeline] {

[Pipeline] stage

[Pipeline] { (Build)

[Pipeline] tool

[Pipeline] envVarsForTool

[Pipeline] withEnv

[Pipeline] {

[Pipeline] sh

+ mvn clean compile

[INFO] Scanning for projects...

[INFO]

-----< com.example:pipeline-ci-demo >-----

[INFO] Building pipeline-ci-demo 1.0.0-SNAPSHOT

[INFO] from pom.xml

[INFO] -----[jar]-----

[INFO]

[INFO] --- clean:3.2.0:clean (default-clean) @ pipeline-ci-demo ---

[INFO] Deleting /Users/sneha/.jenkins/workspace/Jenkins-Maven-Project-1.2/target

[INFO]

[INFO] --- resources:3.4.0:resources (default-resources) @ pipeline-ci-demo ---

[INFO] skip non existing resourceDirectory /Users/sneha/.jenkins/workspace/Jenkins-Maven-Project-1.2/src/main/resources

[INFO]

Jenkins / Jenkins-Maven-Project-1.2 / #2

```
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 0.407 s
[INFO] Finished at: 2026-09-23T16:34:14+05:30
[INFO] -----
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] }

[Integration Tests] [INFO] No tests to run.
[INFO]
[INFO] --- jar:3.4.2:jar (default-jar) @ pipeline-ci-demo ---
[INFO] Building jar: /Users/snehal/.jenkins/workspace/Jenkins-Maven-Project-1.2/target/pipeline-ci-demo-1.0.0-SNAPSHOT.jar
[INFO] -----

[Code Analysis] [INFO] BU[INFO]
[INFO] -----< com.example.pipeline-ci-demo >-----
[INFO] Building pipeline-ci-demo 1.0.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- pmd:3.28.0:pmd (default-cli) @ pipeline-ci-demo ---

[Integration Tests] [INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 0.538 s
[INFO] Finished at: 2026-09-23T16:34:14+05:30
[INFO] -----
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] }

[Code Analysis] [WARNING] Unable to locate Source XRef to link to -- DISABLED
[INFO] PMD version: 7.17.0
[INFO] Rendering content with org.apache.maven.skis:maven-fluido-skin:jar:2.0.0-M9 skin
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 1.526 s
[INFO] Finished at: 2026-09-23T16:34:15+05:30
[INFO] -----
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // parallel
[Pipeline] }
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Project 1.3-Jenkins Master-Agent Distributed build architecture.

Task 1. Install Jenkins Controller

Using my mac as Jenkins controller : Verified access.

S	W	Name	Last Success	Last Failure	Last Duration	# Issues
✓	☁	Jenkins-Maven-Project-1.1	53 min #7	1 hr 3 min #6	17 sec	-
✓	☀	Jenkins-Maven-Project-1.2	21 min #2	N/A	12 sec	-
✓	☁	Maven-pipeline	1 mo 2 days #19	1 mo 2 days #18	12 sec	-
🔑	☀	MultiBranchDemo	1 mo 1 day log	N/A	2.1 sec	-
🔑	☀	myfirstjob	N/A	N/A	N/A	-
✓	☀	myfirstjob1	4 days 23 hr #1	N/A	0.59 sec	-
✓	☀	parameter-pipeline-check	1 mo 2 days #14	1 mo 2 days #7	0.42 sec	-
✓	☁	sonarqube-pipeline	3 mo 13 days #10	3 mo 13 days #9	1 min 34 sec	-
✓	☀	unitTestUsingMaven	3 mo 19 days #2	N/A	4.1 sec	-

Task 2: Setup Jenkins Agent Machine

Creating EC2 Instance as another agent machine.

Instance summary for i-02a15fe85a45a918f (jenkins-agent)

Updated less than a minute ago

Instance ID
i-02a15fe85a45a918f

IPv6 address
-

Hostname type
IP name: ip-172-31-16-34.ec2.internal

Answer private resource DNS name
IPV4 (A)

Auto-assigned IP address
100.59.195.244 (Public IP)

IAM role
-

IMDSv2
Required

Operator
-

Public IPv4 address
100.59.195.244 | [open address ↗](#)

Instance state
Running

Private IP DNS name (IPv4 only)
ip-172-31-16-34.ec2.internal

Instance type
c7i-flex.large

VPC ID
vpc-0448199f99a4c3ef2 | [↗](#)

Subnet ID
subnet-0b2531a6bd847d46 | [↗](#)

Instance ARN
arn:aws:ec2:us-east-1:307555122680:instance/i-02a15fe85a45a918f

Private IPv4 addresses
172.31.16.34

Public DNS
ec2-100-59-195-244.compute-1.amazonaws.com | [open address ↗](#)

Elastic IP addresses
-

AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. | [Learn more ↗](#)

Auto Scaling Group name
-

Managed
false

▼ Instance details

AMI ID
ami-0b6d9d3d33ba97d99

AMI name
ubuntu/images/hvm-ssd-gp3/ubuntu-resolute-26.04-amd64-server-20260604

Stop protection
Disabled

Instance reboot migration
Default (On)

Stop-hibernate behavior
Disabled

State transition reason
-

Monitoring
disabled

Allowed image
-

Launch time
Wed Sep 25 2026 19:09:44 GMT+0530 (India Standard Time) (less than a minute)

Instance auto-recovery
Default

AMI Launch index
0

Credit specification
Not supported by instance type

Platform details
Linux/UNIX

Termination protection
Disabled

AMI location
amazon/ubuntu/images/hvm-ssd-gp3/ubuntu-resolute-26.04-amd64-server-20260604

Lifecycle
normal

Key pair assigned at launch
22sept

Kernel ID
-

Install : Java, Git , maven

Execute below commands :

#apt update

Java Installation :

Task 3: Add New Node in Jenkins

Navigate in Jenkins:

Manage Jenkins → Mana

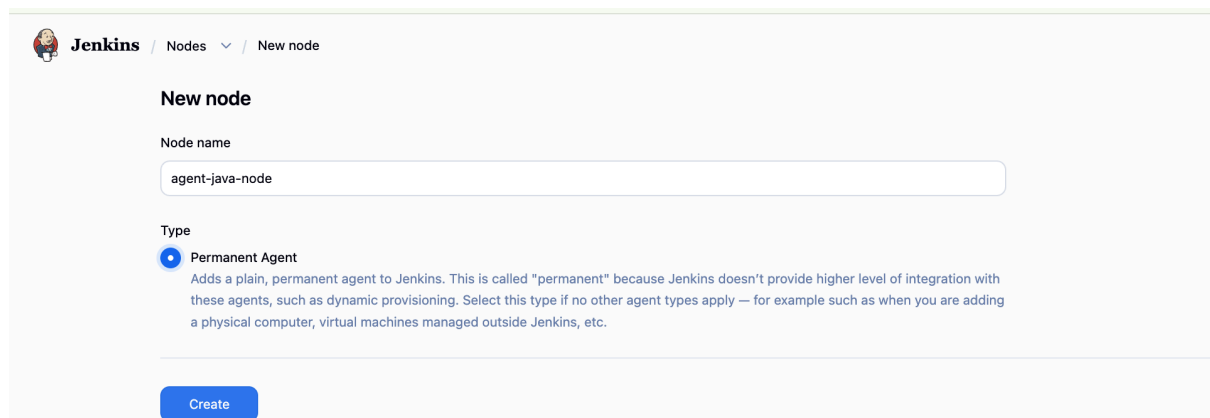
Configure:

Node Name:

agent-java-node

Node Type:

Permanent Agent

The screenshot shows the Jenkins web interface for adding a new node. At the top, there's a breadcrumb trail: 'Jenkins / Nodes / New node'. The main heading is 'New node'. Below it, there's a 'Node name' field with the text 'agent-java-node' entered. Under the 'Type' section, the 'Permanent Agent' option is selected with a radio button. A descriptive text below the radio button explains that this type adds a plain, permanent agent to Jenkins, which is called 'permanent' because it doesn't provide a higher level of integration with other agent types like dynamic provisioning. At the bottom, there is a blue 'Create' button.

Task 4: Configure Agent Settings

For this make sure you create /home/jenkins/agent directory and modify the permissions accordingly.

chmod -R 777 /home/jenkins/agent

Specify:

Remote Root Directory : /home/jenkins/agent

Labels

java-agent

Launch Method

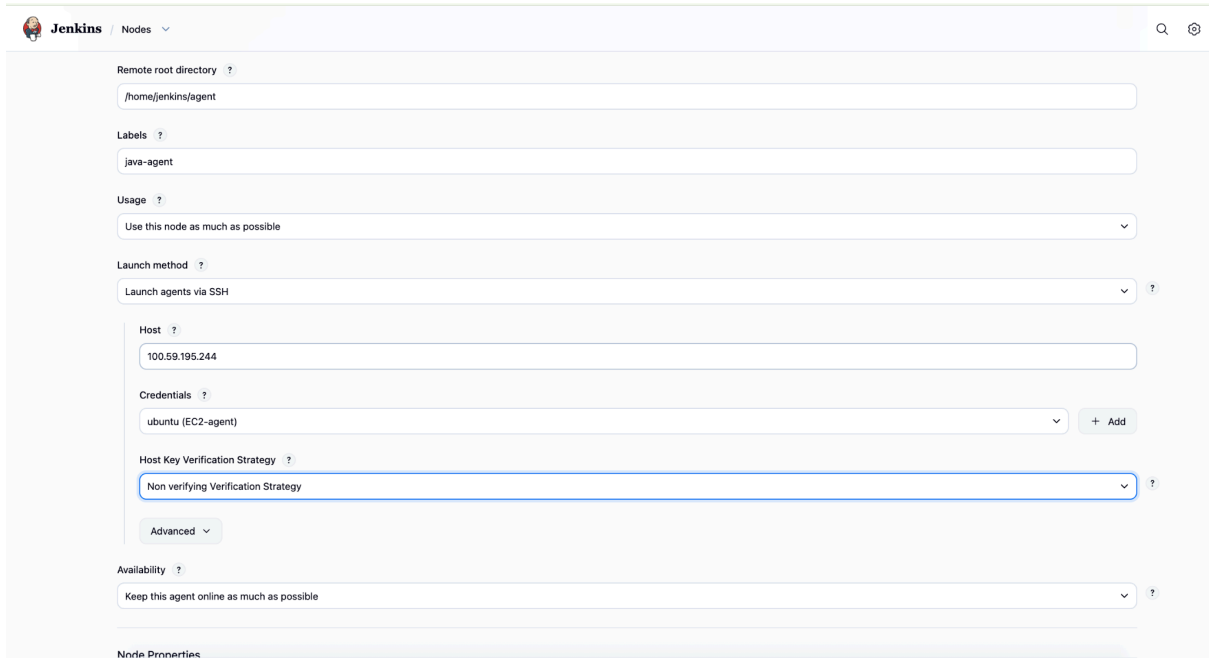
Launch agent via SSH

Task 5: Connect Agent to Controller

Provide: Agent machine IP address

SSH credentials

Once connected, Jenkins will show the agent as online.



The screenshot shows the Jenkins 'New Node' configuration page. The top navigation bar includes the Jenkins logo, the text 'Jenkins', and a 'Nodes' dropdown menu. On the right side of the bar are search and settings icons. The main configuration area contains several fields: 'Remote root directory' with the value '/home/jenkins/agent'; 'Labels' with the value 'java-agent'; 'Usage' with a dropdown set to 'Use this node as much as possible'; 'Launch method' with a dropdown set to 'Launch agents via SSH'; 'Host' with the value '100.59.195.244'; 'Credentials' with a dropdown set to 'ubuntu (EC2-agent)' and an '+ Add' button; 'Host Key Verification Strategy' with a dropdown set to 'Non verifying Verification Strategy'; an 'Advanced' dropdown button; and 'Availability' with a dropdown set to 'Keep this agent online as much as possible'. At the bottom, there is a 'Node Properties' section which is currently collapsed.

Jenkins Nodes

Remote root directory ?
/home/jenkins/agent

Labels ?
java-agent

Usage ?
Use this node as much as possible

Launch method ?
Launch agents via SSH

Host ?
100.59.195.244

Credentials ?
ubuntu (EC2-agent) + Add

Host Key Verification Strategy ?
Non verifying Verification Strategy

Advanced

Availability ?
Keep this agent online as much as possible

Node Properties

Jenkins

Manage Jenkins

Credentials

System

Global

ec2-user (EC2-agent)

ec2-user (EC2-agent)

EC2-agent

Usage

This credential has not been recorded as used anywhere.

Note: usage tracking requires the cooperation of plugins and consequently may not be available.

Update credential

Scope ?

Global (Jenkins, nodes, items, all child items, etc)

ID ?

EC2-agent-key

Description ?

EC2-agent

Username

ubuntu

☐ Treat username as secret ?

Private Key

☒ Enter directly

Key

Concealed for Confidentiality

Replace

Passphrase

Save

Jenkins

Nodes

Remote root directory ?

/home/jenkins/agent

Labels ?

java-agent

Usage ?

Use this node as much as possible

Launch method ?

Launch agents via SSH

Host ?

100.59.195.244

Credentials ?

ubuntu (EC2-agent)

+ Add

Host Key Verification Strategy ?

Non verifying Verification Strategy

Advanced

Availability ?

Keep this agent online as much as possible

Node Properties

JenkinsNodesagent-java-nodeLog

```
Pipes (A) US= ({}= {} )
PPID=2918
PS4=' '
PWD=/home/ubuntu
SHELL=/bin/bash
SHELLOPTS=braceexpand:hashall:interactive-comments
SHLVL=1
SSH_CLIENT='58.84.61.7 15084 22'
SSH_CONNECTION='58.84.61.7 15084 172.31.16.34 22'
TERM=dumb
UID=1000
USER=ubuntu
XDG_RUNTIME_DIR=/run/user/1000
XDG_SESSION_CLASS=user
XDG_SESSION_ID=3
XDG_SESSION_TYPE=tty
_='|'
[09/23/26 19:14:57] [SSH] Starting sftp client.
[09/23/26 19:14:59] [SSH] Copying latest remoting.jar...
[09/23/26 19:15:21] [SSH] Copied 1,407,938 bytes.
Expanded the channel window size to 4MB
[09/23/26 19:15:22] [SSH] Starting agent process: cd "/home/jenkins/agent" && java -jar remoting.jar -workDir /home/jenkins/agent -jar-cache
/home/jenkins/agent/remoting/jarCache
Sep 23, 2026 1:45:22 PM org.jenkinsci.remoting.engine.WorkDirManager initializeWorkDir
INFO: Using /home/jenkins/agent/remoting as a remoting work directory
Sep 23, 2026 1:45:22 PM org.jenkinsci.remoting.engine.WorkDirManager setupLogging
INFO: Both error and output logs will be printed to /home/jenkins/agent/remoting
<====[JENKINS REMOTING CAPACITY]====>channel started
Remoting version: 3355.3357.v931d3c992987
Launcher: SSHTLSLauncher
Communication Protocol: Standard in/out
This is a Unix agent
Agent successfully connected and online
```

Task 6: Assign Jobs to Agent

JenkinsAllNew Item

New Item

Enter an item name

Jenkins-Maven-Project-1.3

Select an item type

Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.

Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

Folder

Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.

Multibranch Pipeline

Creates a set of Pipeline projects according to detected branches in one SCM repository.

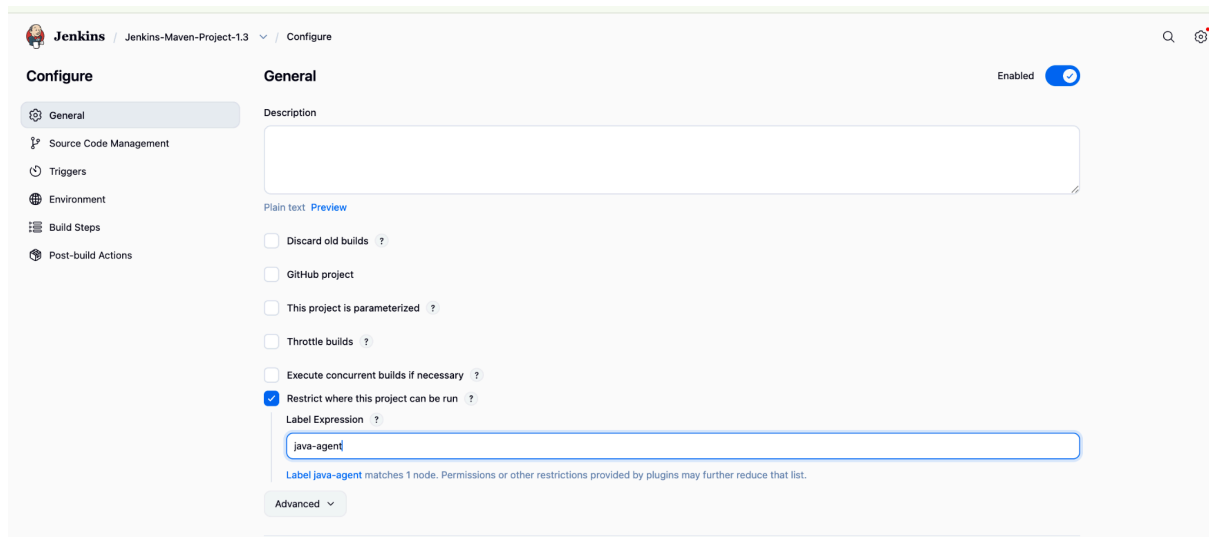
Organization Folder

Creates a set of multibranch project subfolders by scanning for repositories.

Duplicate an existing item

Create a new item by copying the configuration of an existing one.

OK



Build Steps → Add build step → Execute shell
echo "Running on Jenkins Java Agent"

```
echo "Hostname:"  
hostname
```

```
echo "Java version:"  
java -version
```

```
echo "Java location:"  
which java
```

```
echo "Jenkins workspace:"  
pwd
```

Jenkins / Jenkins-Maven-Project-1.3 / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment**
- Build Steps
- Post-build Actions

Environment

Configure settings and variables that define the context in which your build runs, like credentials, paths, and global parameters.

- ☐ Delete workspace before build starts
- ☐ Use secret text(s) or file(s) ?
- ☐ Add timestamps to the Console Output
- ☐ Inspect build log for published build scans
- ☐ Prepare SonarQube Scanner environment ?
- ☐ Terminate a build if it's stuck
- ☐ With Ant ?

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute shell ?

Command

[See the list of available environment variables](#)

```
echo "Running on Jenkins Java Agent"
echo "Hostname:"
hostname
```

Advanced ▾

+ Add build step

Save Apply

Jenkins / Jenkins-Maven-Project-1.3 / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps**
- Post-build Actions

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute shell ?

Command

[See the list of available environment variables](#)

```
echo "Java location:"
which java
echo "Jenkins workspace:"
pwd
```

Advanced ▾

+ Add build step

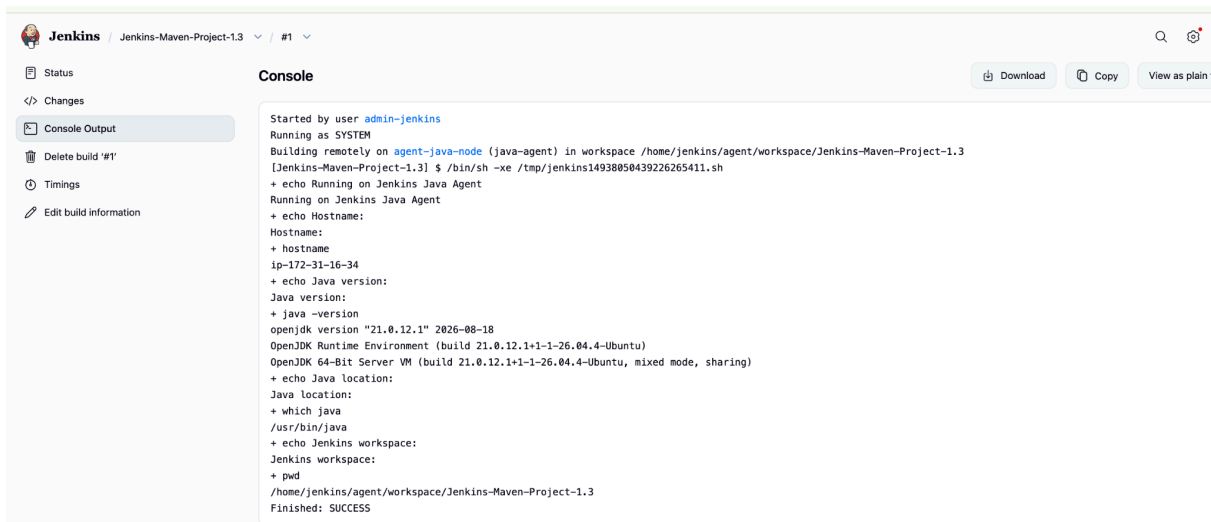
Post-build Actions

Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.

+ Add post-build action

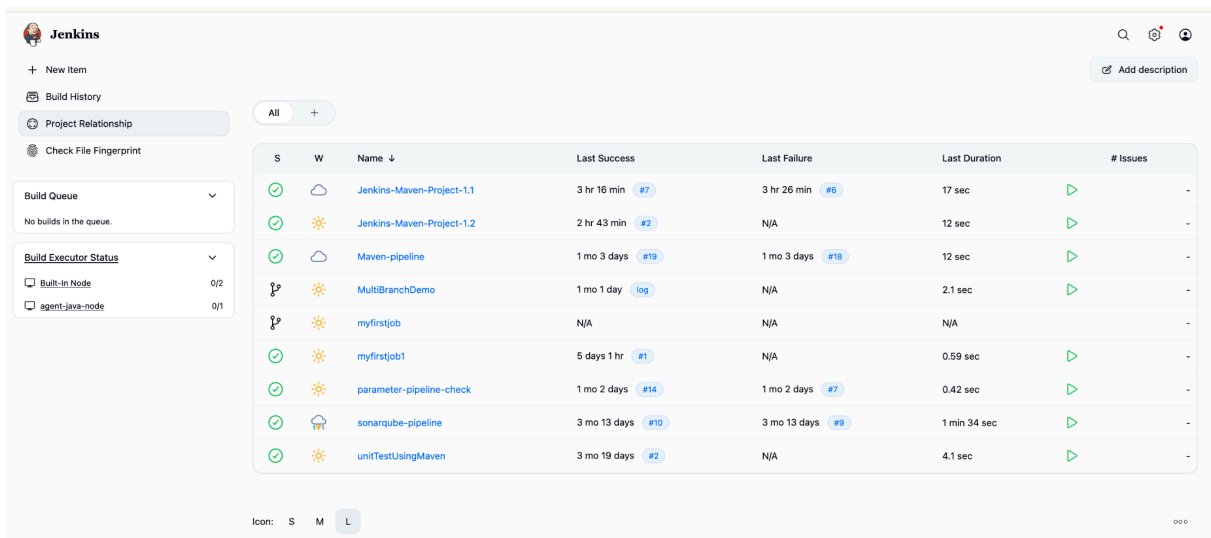
Save Apply

Save and apply and then execute



Deliverables :

-Screenshot of jenkins controller dashboard.



-Screenshot of configured Jenkins Agent .

Jenkins / Nodes / agent-java-node / Log

Status
Delete Agent
Configure
Build History
Load Statistics
Script Console
Log
System Information
Disconnect

Build Executor Status 0/1

```

SSHLauncher(host='100.59.195.244', port=22, credentialsId='EC2-agent-key', jvmOptions='', javaPath='', prefixStartSlaveCmd='', suffixStartSlaveCmd='',
launchTimeoutSeconds=60, maxNumRetries=10, retryWaitTime=15,
sshHostKeyVerificationStrategy=hudson.plugins.sshslaves.verifiers.NonVerifyingKeyVerificationStrategy, tcpNoDelay=true, trackCredentials=true)
[09/23/26 19:14:54] [SSH] Opening SSH connection to 100.59.195.244:22.
[09/23/26 19:14:55] [SSH] WARNING: SSH Host Keys are not being verified. Man-in-the-middle attacks may be possible against this connection.
[09/23/26 19:14:56] [SSH] Authentication successful.
[09/23/26 19:14:57] [SSH] The remote user's environment is:
BASH=/usr/bin/bash
BASHOPTS=checkwinsize:cmdhist:complete_fullquote:extquote:force_ignores:globasciiranges:globskipdots:hostcomplete:interactive_comments:patsub_replacement:progcomp
:promptvars:sourcepath
BASH_ALIASES=()
BASH_ARGC=(0)
BASH_ARGV=()
BASH_CMDS=()
BASH_EXECUTION_STRING=set
BASH_LINENO=()
BASH_LOADABLES_PATH=/usr/local/lib/bash:/usr/lib/bash:/opt/local/lib/bash:/usr/pkg/lib/bash:/opt/pkg/lib/bash:.
BASH_SOURCE=()
BASH_VERSION=([0]="5" [1]="3" [2]="9" [3]="1" [4]="release" [5]="x86_64-pc-linux-gnu")
BASH_VERSION=5.3.9(1)-release
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
DIRSTACK=()
EUID=1000
GROUPS=()
HOME=/home/ubuntu
HOSTNAME=ip-172-31-16-34
HOSTTYPE=x86_64
IFS=$'\n'
LANG=C.UTF-8
LOGNAME=ubuntu
MACHTYPE=x86_64-pc-linux-gnu

```

-Screenshot of successful build executed on agent

Jenkins / Jenkins-Maven-Project-1.3

Status ✔ Jenkins-Maven-Project-1.3

Changes
Workspace
Build Now
Delete Project
Configure
Rename

Permalinks

- Last build (#1), 42 sec ago
- Last stable build (#1), 42 sec ago
- Last successful build (#1), 42 sec ago
- Last completed build (#1), 42 sec ago

Builds > Filter /

Today

✔ #1 7:23 PM

-Console output logs.

Jenkins / Jenkins-Maven-Project-1.3 / #1

Status
Changes
Console Output
Timings
Edit build information

Console

Download Copy View as plain text

```

Started by user admin-jenkins
Running as SYSTEM
Building remotely on agent-java-node (java-agent) in workspace /home/jenkins/agent/workspace/Jenkins-Maven-Project-1.3
[Jenkins-Maven-Project-1.3] $ /bin/sh -xe /tmp/jenkins14938050439226265411.sh
+ echo Running on Jenkins Java Agent
Running on Jenkins Java Agent
+ echo Hostname:
Hostname:
+ hostname
ip-172-31-16-34
+ echo Java version:
Java version:
+ java -version
openjdk version "21.0.12.1" 2026-08-18
OpenJDK Runtime Environment (build 21.0.12.1-1-26.04.4-Ubuntu)
OpenJDK 64-Bit Server VM (build 21.0.12.1-1-26.04.4-Ubuntu, mixed mode, sharing)
+ echo Java location:
Java location:
+ which java
/usr/bin/java
+ echo Jenkins workspace:
Jenkins workspace:
+ pwd
/home/jenkins/agent/workspace/Jenkins-Maven-Project-1.3
Finished: SUCCESS

```

Project 1.4 - Jenkins + SonarQube Quality Gate Pipeline.

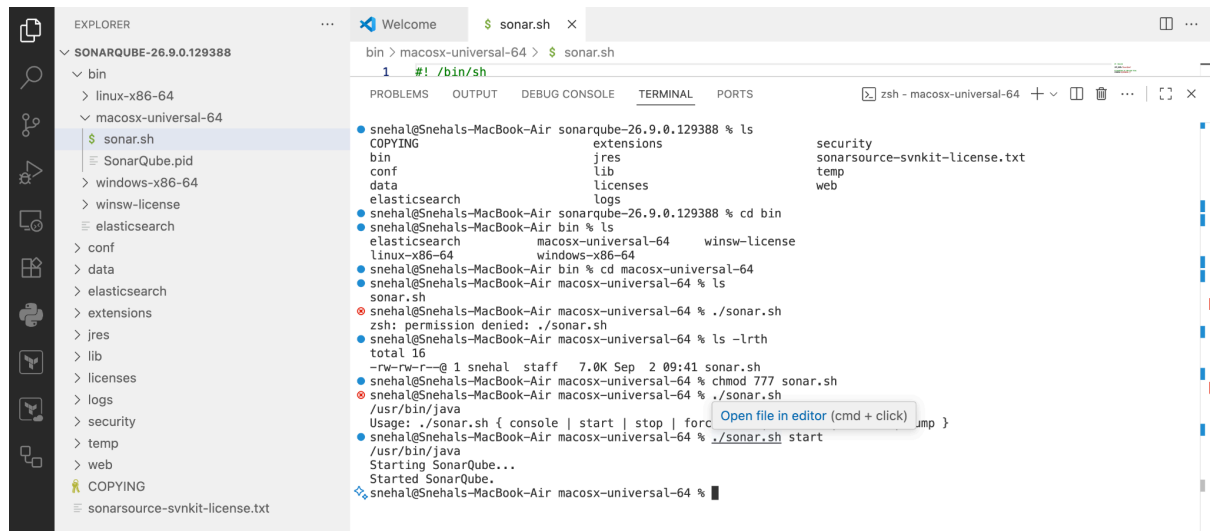
Task1. Install SonarQube.

Run SonarQube Server.

Access dashboard: <http://localhost:9000>

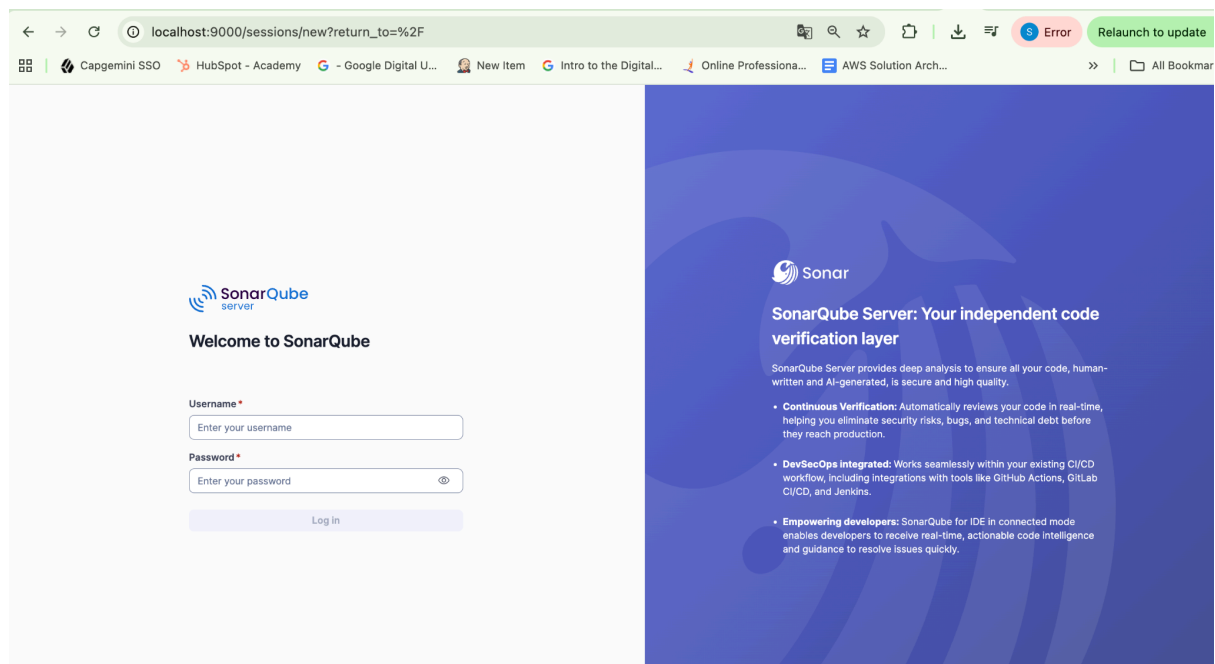
IMP : You need java version 21 installed

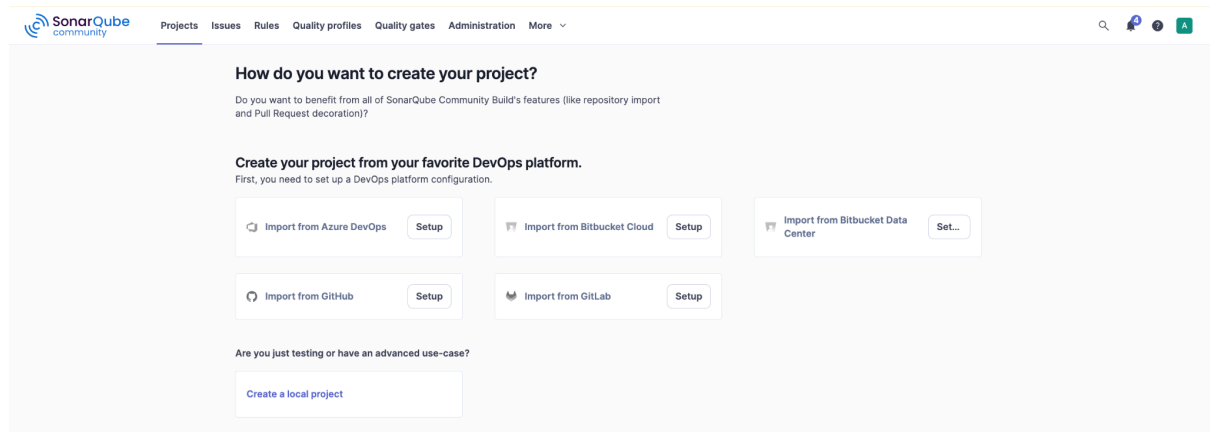
Download the community version :



```
bin > macosx-universal-64 > $ sonar.sh
1  #! /bin/sh
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
zsh - macosx-universal-64

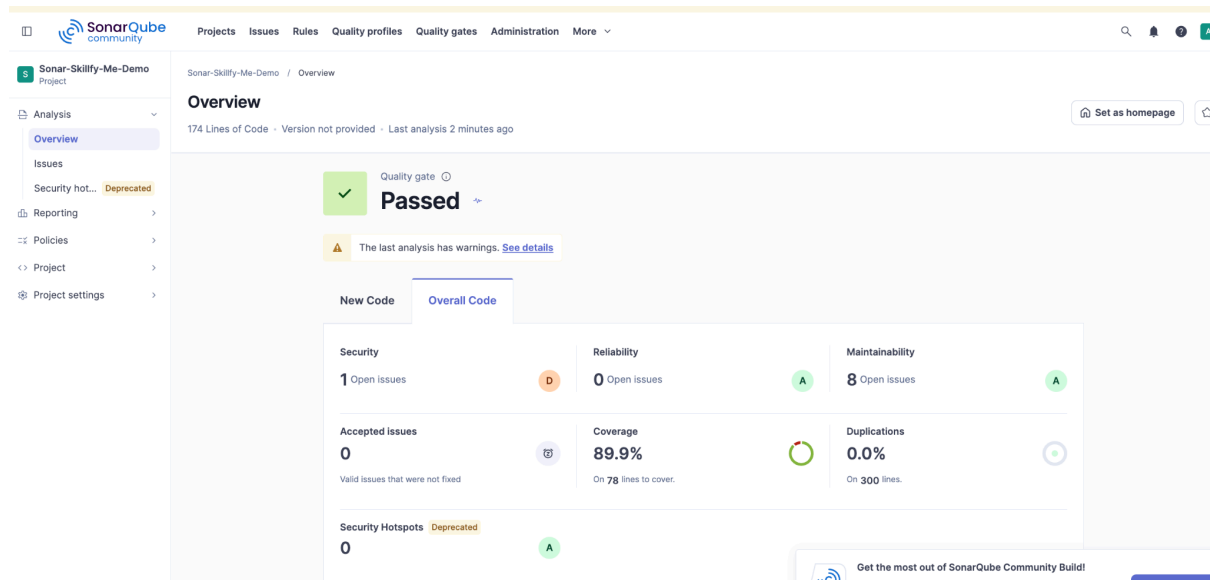
• snehal@Snehals-MacBook-Air sonarqube-26.9.0.129388 % ls
COPYING extensions security
bin jres sonarsource-svnkit-license.txt
conf lib temp
data licenses web
elasticsearch logs
• snehal@Snehals-MacBook-Air sonarqube-26.9.0.129388 % cd bin
• snehal@Snehals-MacBook-Air bin % ls
elasticsearch macosx-universal-64 winsw-license
linux-x86-64 windows-x86-64
• snehal@Snehals-MacBook-Air bin % cd macosx-universal-64
• snehal@Snehals-MacBook-Air macosx-universal-64 % ls
sonar.sh
• snehal@Snehals-MacBook-Air macosx-universal-64 % ./sonar.sh
zsh: permission denied: ./sonar.sh
• snehal@Snehals-MacBook-Air macosx-universal-64 % ls -lrth
total 16
-rw-rw-r--@ 1 snehal staff 7.0K Sep 2 09:41 sonar.sh
• snehal@Snehals-MacBook-Air macosx-universal-64 % chmod 777 sonar.sh
• snehal@Snehals-MacBook-Air macosx-universal-64 % ./sonar.sh
/usr/bin/java
Usage: ./sonar.sh { console | start | stop | force | help }
• snehal@Snehals-MacBook-Air macosx-universal-64 % ./sonar.sh start
Starting SonarQube...
Started SonarQube.
• snehal@Snehals-MacBook-Air macosx-universal-64 %
```





Login and create local project :

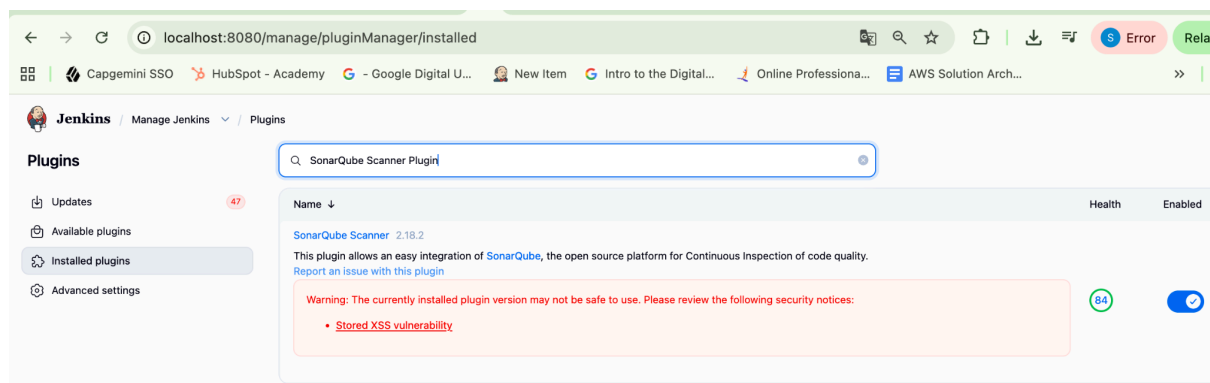
I have created [Sonar-Skillfy-Me-Demo](#)



Task2: Install Jenkins Sonarqube plugin

Navigate to manage jenkins→Manage Plugin

Install SonarQube Scanner Plugin

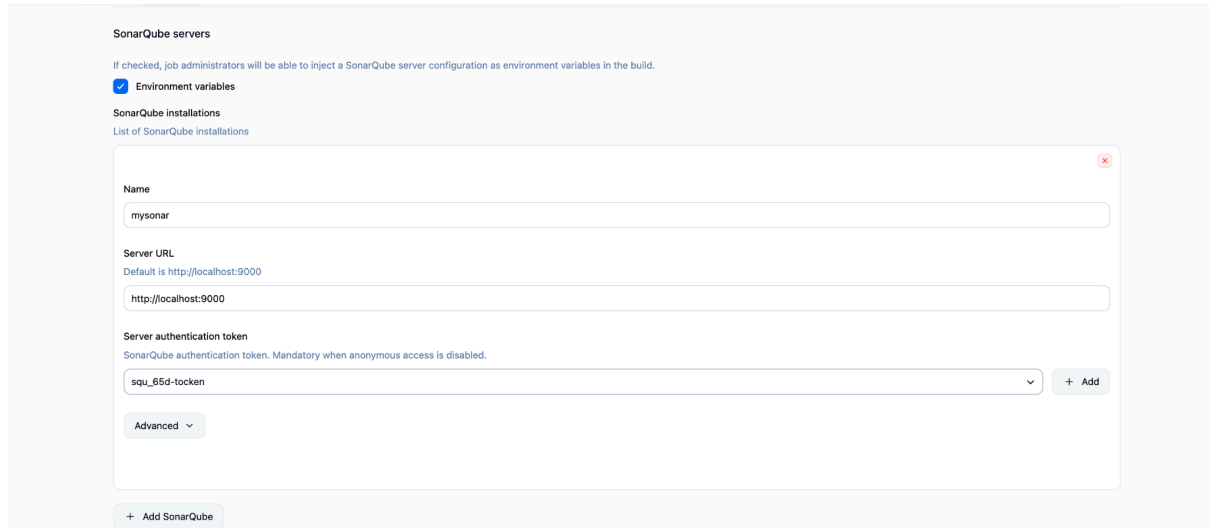


Task3: Configure Sonarqube server in jenkins

Navigate to: Manage Jenkins→configure System.

Add SonarQube Server: SonarQube server URL.

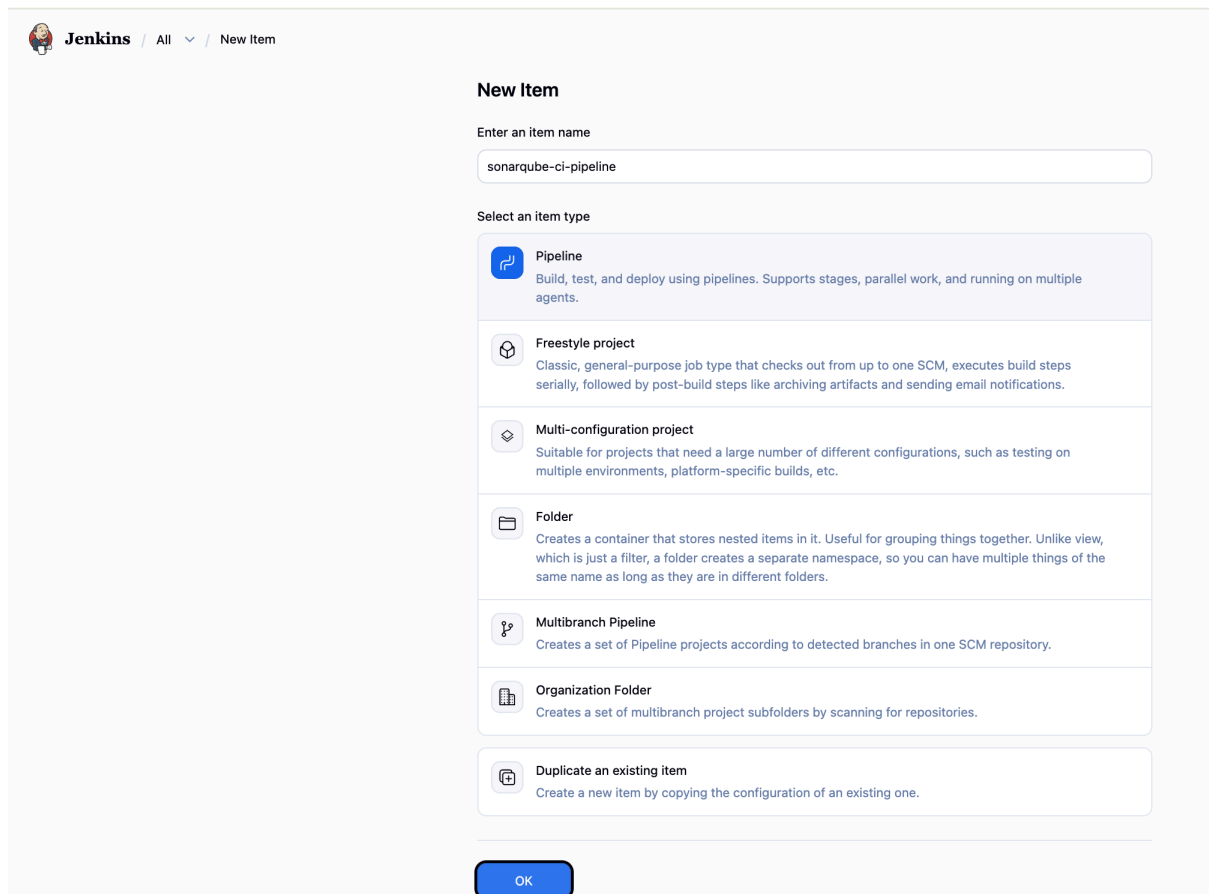
Authentication token.



The screenshot shows the 'SonarQube servers' configuration page in Jenkins. At the top, there's a section 'SonarQube servers' with a checkbox 'Environment variables' checked. Below this is a section 'SonarQube installations' with a sub-header 'List of SonarQube installations'. A form is displayed with three main fields: 'Name' (containing 'mysonar'), 'Server URL' (containing 'http://localhost:9000'), and 'Server authentication token' (containing 'squ_65d-token'). There are also 'Advanced' and 'Add' buttons at the bottom of the form.

Task 4: Create Jenkins Pipeline Job

Create a new pipeline job: sonarqube-ci-pipeline



The screenshot shows the 'New Item' page in Jenkins. The page has a header with the Jenkins logo and 'All / New Item'. The main section is titled 'New Item' and contains a form to create a new item. The 'Enter an item name' field contains 'sonarqube-ci-pipeline'. Below this is a section 'Select an item type' with a list of options: 'Pipeline', 'Freestyle project', 'Multi-configuration project', 'Folder', 'Multibranch Pipeline', 'Organization Folder', and 'Duplicate an existing item'. The 'Pipeline' option is selected. At the bottom of the page is an 'OK' button.

Task 5: Create Jenkinsfile

Using Below Jenkinsfile for demo purpose :

```
pipeline{
    agent any
    tools{
        maven 'maven3'
    }
    stages{
    stage('checkout Code')
    {
        steps{
            git 'https://github.com/demo1orgtoday/SonarQubeCoverageJava.git'
        }
    }

    stage('Build Code')
    {
        steps{
            sh 'mvn package'
        }
    }
    stage('SonarScan'){
        environment{
            scannerHome = tool 'sonartool'
        }
        steps{
withSonarQubeEnv('mysonar'){

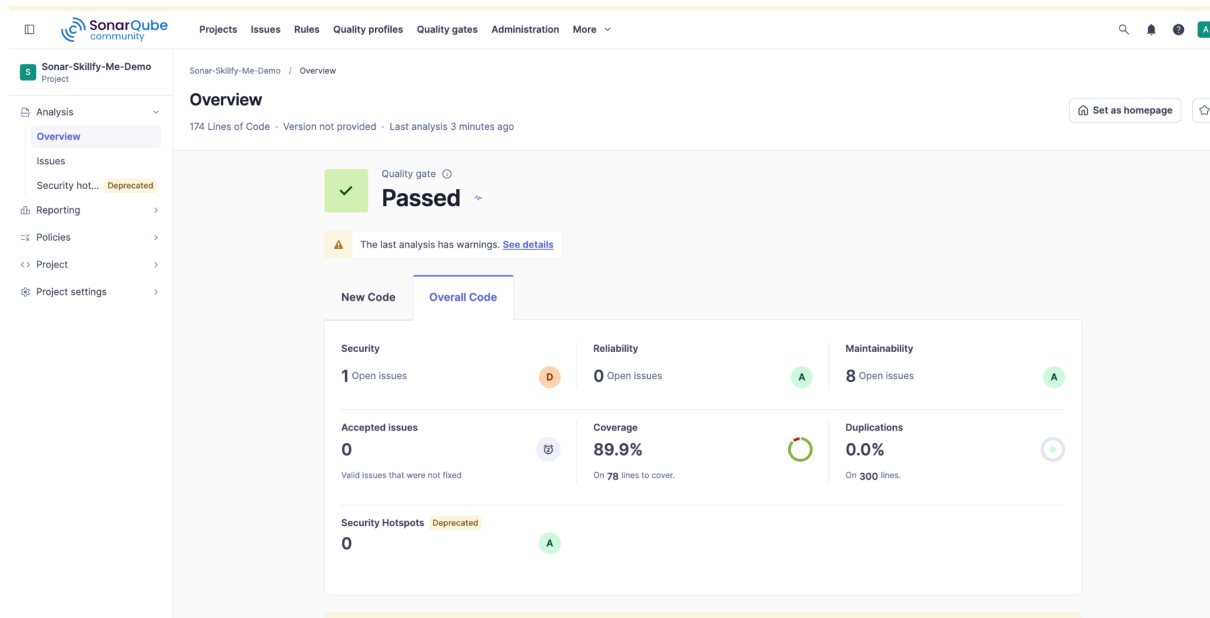
            sh "${scannerHome}/bin/sonar-scanner \
              -Dsonar.projectKey=Sonar-Skillfy-Me-Demo \
              -Dsonar.java.binaries=target/classes \

            ""
        }

    }

    }
    }
}
```

Task 6: Verify Sonarqube Analysis

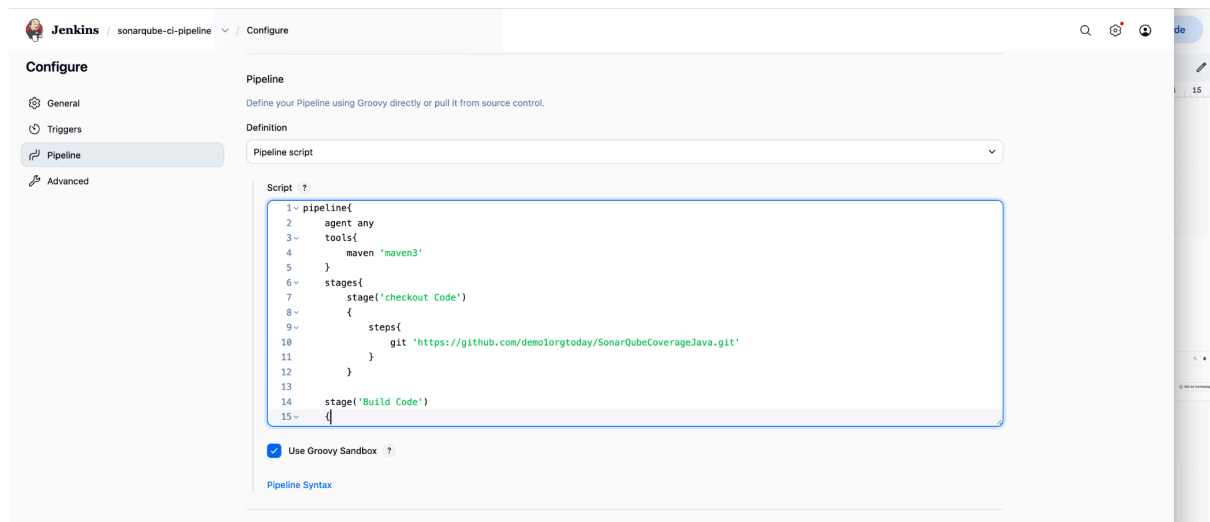


Deliverables :

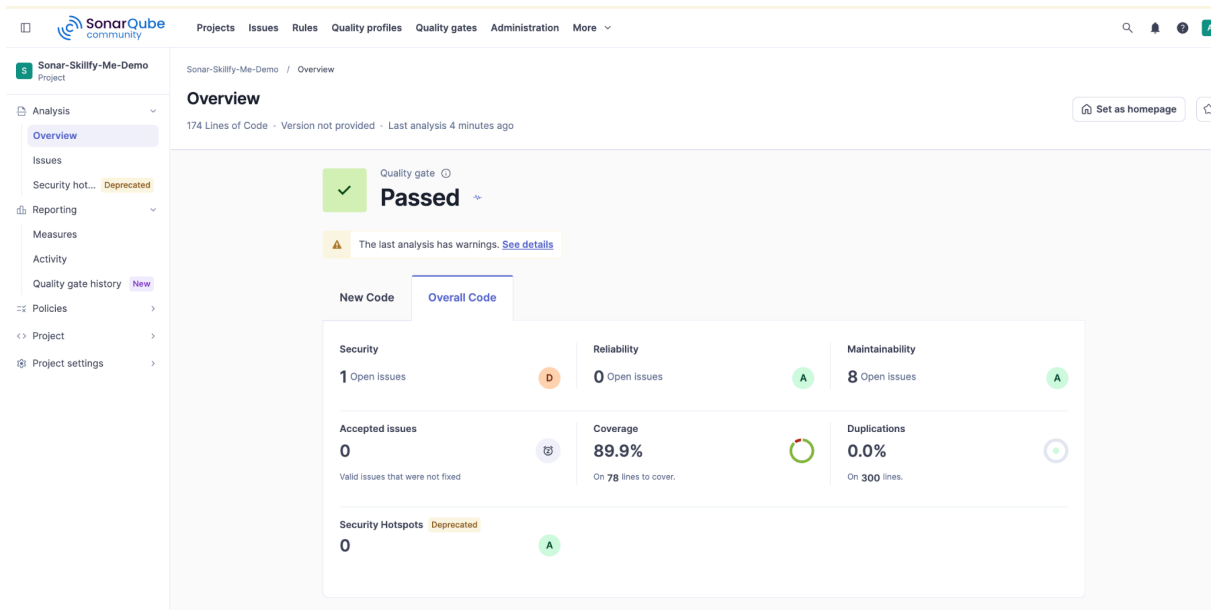
-Github Repository link :

'<https://github.com/demo1orgtoday/SonarQubeCoverageJava.git>'

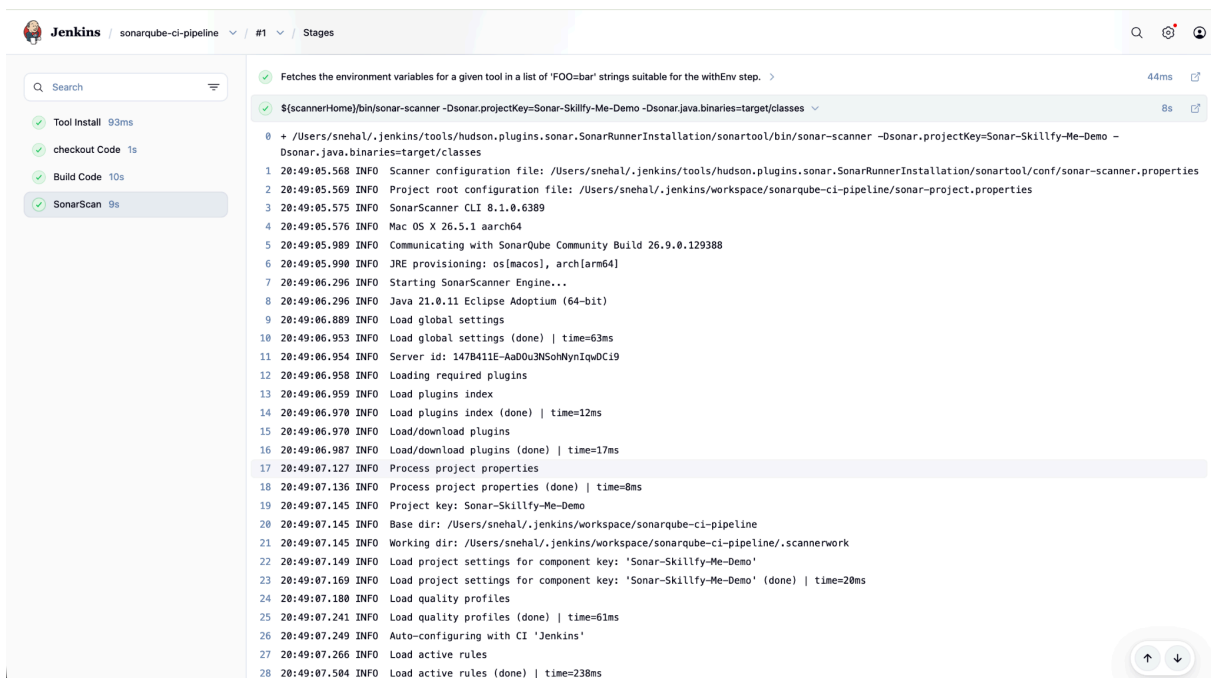
-Screenshot of jenkins pipeline configuration



-Screenshot of SonarQube Analysis report.



-Screenshot of jenkins console output.





```
20:49:12.293 INFO Sensor Zero Coverage Sensor
20:49:12.293 INFO Sensor Zero Coverage Sensor (done) | time=0ms
20:49:12.293 INFO Sensor Java CPD Block Indexer
20:49:12.303 INFO Sensor Java CPD Block Indexer (done) | time=9ms
20:49:12.303 INFO ----- Gather SCA dependencies on project
20:49:12.309 INFO Dependency analysis skipped
20:49:12.311 INFO SCM Publisher SCM provider for this project is: git
20:49:12.311 INFO SCM Publisher 11 source files to be analyzed
20:49:12.854 INFO SCM Publisher 11/11 source files have been analyzed (done) | time=543ms
20:49:12.857 INFO CPD Executor 1 file had no CPD blocks
20:49:12.858 INFO CPD Executor Calculating CPD for 2 files
20:49:12.860 INFO CPD Executor CPD calculation finished (done) | time=3ms
20:49:12.866 INFO SCM revision ID '6823daafc63ff3ce6206169afa3937effc000d9c'
20:49:12.964 INFO Analysis report generated in 99ms, dir size=340.5 kB
20:49:12.988 INFO Analysis report compressed in 24ms, zip size=58.9 kB
20:49:13.005 INFO Analysis report uploaded in 17ms
20:49:13.006 INFO ANALYSIS SUCCESSFUL, you can find the results at: http://localhost:9000/dashboard?id=Sonar-Skillfy-Me-Demo
20:49:13.006 INFO Note that you will be able to access the updated dashboard once the server has processed the submitted analysis report
20:49:13.006 INFO More about the report processing at http://localhost:9000/api/ce/task?id=4ef91d5c-4ec8-40f2-818c-6e1c5713b394
20:49:13.096 INFO Analysis total time: 6.087 s
20:49:13.098 INFO SonarScanner Engine completed successfully
20:49:13.112 INFO EXECUTION SUCCESS
20:49:13.113 INFO Total time: 7.559s
[Pipeline] }
[Pipeline] // withSonarQubeEnv
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```